

Digital Electronics Laboratory

Laboratory Exercise 1. Logic gates analysis

Objectives:

- To investigate gates operation.
- To study some fundamental laws of Boolean Algebra.
- To become familiar with logic circuits.

Introduction

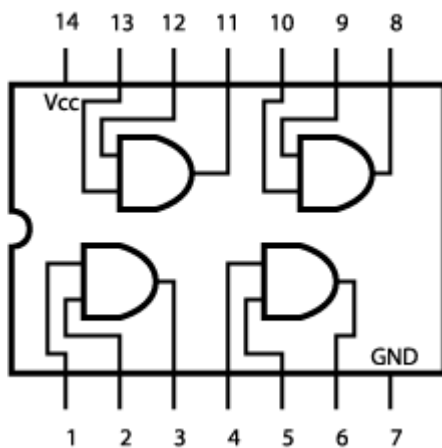
This is the introductory laboratory session, to allow you to become familiar with very basic digital circuits.

Pre-Laboratory

There are several tasks that you must perform **prior** to sitting this laboratory:

1. Read the laboratory assignment in full.
2. Draw up the truth tables for a 2-input AND and a 2-input OR gate.
3. Explain the Boolean algebra Associative, Commutative and Distributive Laws.
4. Derive a Truth-Table for the function $F=AB+CD$.

Useful Chip Diagrams:



7408(AND)

The Laboratory:

Section 1. AND Gate Implementation

(a) Connect one of the 2-input AND gates (74LS08) as shown in Figure 1



Figure 1. The 2-input AND gate.

(b) Vary the inputs A and B to obtain all the possible combinations and complete the truth table (as below) for the AND gate.

A	B	F	
0	0	0	
0	1	0	
1	0	0	
1	1	1	

(c) The 2-input AND gate can be extended to a 4-input AND gate as shown in Figure 2. Connect the gates as shown and generate the values for the truth table below.

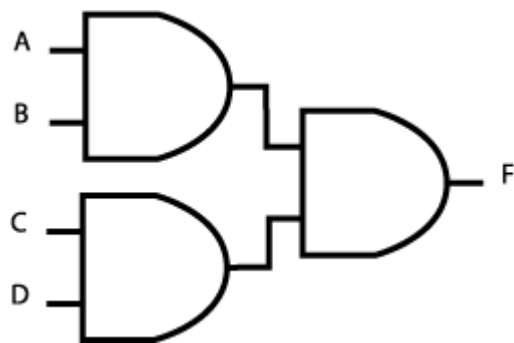


Figure 2. The 4-input AND gate, built using 2-input AND gates.

A	B	C	D	F
0	0	0	0	
0	0	0	1	
0	0	1	0	
0	0	1	1	
0	1	0	0	
0	1	0	1	
0	1	1	0	
0	1	1	1	
1	0	0	0	
1	0	0	1	

1	0	1	0	
1	0	1	1	
1	1	0	0	
1	1	0	1	
1	1	1	0	
1	1	1	1	

(d) Design a 3-input AND gate that is composed of 2-input AND gates. Measure the outputs and generate a truth table. Describe your design. How would you create a 5 input AND gate?

Section 2. Associative and Commutative Laws

(a) Connect AND gates as shown in Figure 3 to implement the Boolean equations $F = A(BC)$ and $G = (AB)C$.

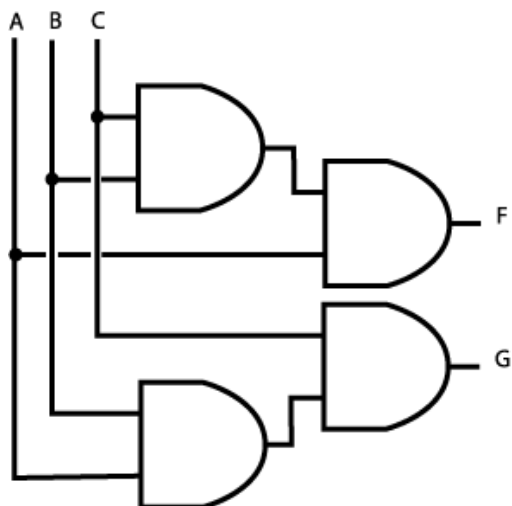


Figure 3. Proving The Associative Law.

(b) Vary inputs A, B, and C to obtain all of the possible combinations and to check the associative law: $A(BC) = (AB)C$. Fill in the truth table below:

A	B	C	F	G
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

(c) How would you demonstrate the Commutative Law using logic gates? Describe your design and implement it using logic gates. Check all the combinations of inputs and the corresponding outputs to insure that it works. Show the resulting truth table.

Section 3. Laws of Boolean Algebra

(a) To demonstrate the Distributive law, connect AND, OR gates as shown in Figure 4. Vary the inputs A, B and C to obtain all possible combinations and check that the outputs F and G are identical.

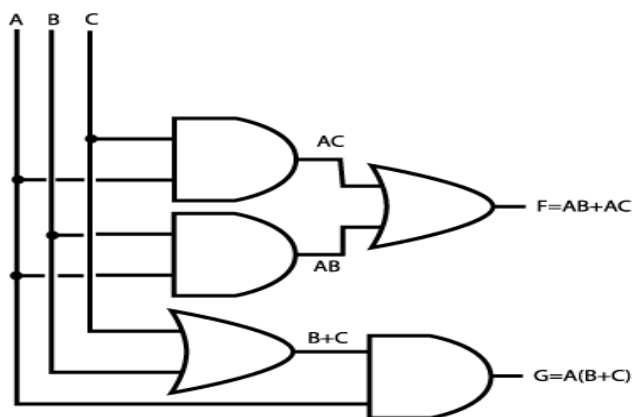


Figure 4. The Distributive Law.

Give the outputs in a truth table as shown below:

A	B	C	F	G
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

Section 4. Disconnected Inputs

(a) Check the behaviour of an OR gate when one of its inputs is open (disconnected - not connected to ground). Check the behaviour when two of the inputs are open. Describe what you observed. Do the same for an AND gate.

Section 5. Boolean Equations

(a) Consider the logic statement: "If Mary obtains permission from her mother or her father and if Joe or Tom pick her up, she may go to the cinema". This statement may be expressed as a Boolean equation using the following Boolean variables:

- F = Mary will go to the cinema (true/false)
- A = Her mother will give her permission (true/false)
- B = Her father will give her permission (true/false)
- C = Joe will pick her up (true/false)
- D = Tom will pick her up (true/false)

Then $F = (A+B).(C+D)$ is the Boolean equation that describes this statement. Implement the circuit for this logic equation and complete the Truth Table below.

A	B	C	D	F
0	0	0	0	
0	0	0	1	
0	0	1	0	
0	0	1	1	
0	1	0	0	
0	1	0	1	
0	1	1	0	
0	1	1	1	
1	0	0	0	
1	0	0	1	
1	0	1	0	
1	0	1	1	
1	1	0	0	
1	1	0	1	
1	1	1	0	
1	1	1	1	

Section 6. Conclusions

State briefly, but clearly, what you have gained from this assignment. Outline aspects that you have noted within the experiment outside of the questions asked.